KOP AI

One Hub. Two Modes. Zero Material Chaos.

01

A PROBLEM WORTH SOLVING

PROBLEM

- Students juggle PDFs, YouTube, screenshots across **multiple platforms**
- **Last-minute learners** need breadth; long-term learners need depth
- Existing tools offer only **one learning pace**
- **No single hub** for all study materials
- AI summaries **lack context** from multiple sources

THE REAL IMPACT

- **3+ hours wasted** weekly switching between disconnected apps and re-finding materials
- Exam anxiety from not knowing what to prioritize or where **knowledge gaps exist**
- **Critical concepts slip** through because they're buried across different sources
- Current AI summarizers give everyone the **same depth** regardless of timeline urgency
- No feedback loop connecting quiz performance back to personalized study recommendations

02

OUR BIG IDEA

QUICK MODE

100-word summaries, 2-3 question quizzes, breadth-first coverage

COMPREHENSIVE MODE

Full study books, deep explanations, focus mode

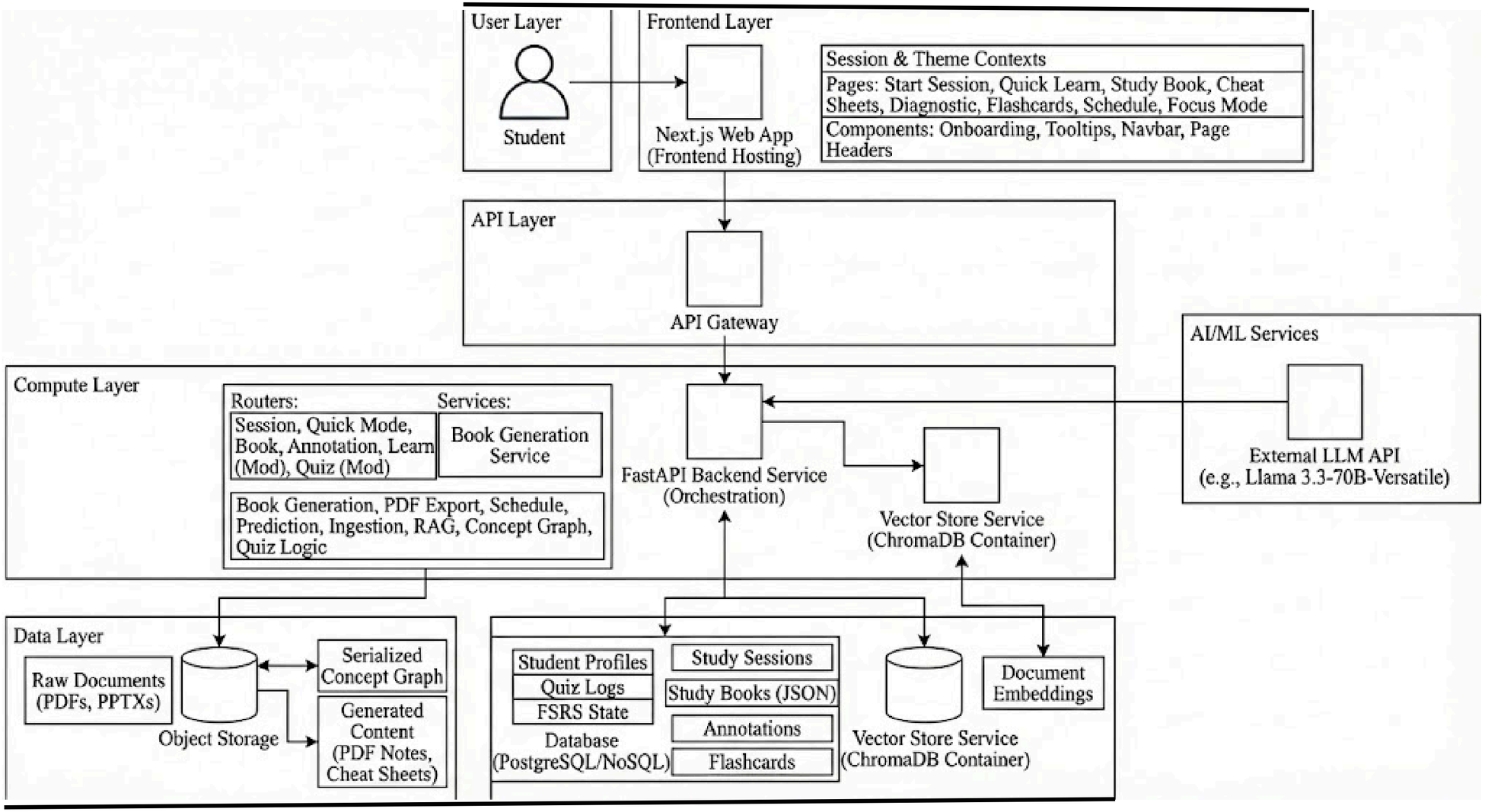
AI STUDY HUB

Both centralize all materials with AI-powered concept extraction

FLOW MODE

Smart switching between modes preserves progress

SYSTEM ARCHITECTURE:



05

CORE FEATURES

INTELLIGENT INGESTION

- Multi-format upload: PDFs, YouTube videos, images with OCR
- Handles complex layouts, tables, mathematical notation
- Auto-tags sources with timestamps for quick reference
- Unified searchable hub with cross-source linking

AI-POWERED LEARNING

- GPT-4 extracts concepts and assigns importance scores
- RAG system generates context-aware summaries
- Mode-adaptive: bullets for Quick, chapters for Comprehensive
- Knowledge graph maps prerequisites in logical order

SMART ASSESSMENT

- Bloom's taxonomy quizzes (Remember → Analyze levels)
- Adaptive difficulty based on mode and performance
- Instant feedback with detailed explanations
- Results feed into mastery tracking and recommendations

PROGRESS INTELLIGENCE

- Hybrid mastery scoring: accuracy × speed × retention
- Visual dashboards track concepts, quizzes, streaks
- Auto-flags weak topics from quiz performance
- Recommends next concepts based on gaps and prerequisites

05

CORE FEATURES

STUDY TOOLS

- Auto-generated cheat sheets and formula cards
- Spaced repetition flashcards with FSRS scheduling
- Study schedule based on exam date and time available
- Session tracking: time spent, concepts covered, patterns

EXPORT & SHARING

- Download PDFs with annotations and highlights preserved
- Citations link back to original sources with page numbers
- Offline access for studying without internet
- Focus mode for distraction-free deep learning

TECH STACK

MODERN, PRODUCTION-READY ARCHITECTURE

FRONTEND

- Next.js 14 (React): Server-side rendering for instant dashboard loads
- Tailwind CSS: Responsive, modern UI design system
- shadcn/ui: Accessible component library for polished UX
- React Query: Optimized data fetching and caching

BACKEND & API

- FastAPI (Python): High-performance async orchestration
- PostgreSQL: Structured data storage with SQLAlchemy ORM
- ChromaDB: Persistent vector database for semantic search
- Redis: Session management and caching layer

AI & INTELLIGENCE LAYER

- OpenAI GPT-4: Summary generation and quiz creation
- LangChain: RAG workflow orchestration and prompt management
- SentenceTransformers: Embedding generation (all-MiniLM-L6-v2)
- NetworkX: Knowledge graph construction and traversal

TWO STUDY MODES:

FEATURE	QUICK MODE	COMPREHENSIVE MODE
Theme	Dark (for late-night study)	Light (with focus mode)
Summary Length	100–150 words, bullet format	300–500 words, study book
Study Tools	Cheat sheets, flashcards	Annotations, highlights, PDFs
Learning Style	Breadth-first, fast coverage	Depth-first, mastery focus
Progress Metric	Concepts skimmed/completed	Chapters mastered

THANK *You!*